

The Terminology of Set Theory

This chapter doesn't cover set theory as completely as a math or computer science textbook would. But it covers enough to justify starting with an explanation of the discipline's basic terminology, as doing so will make the rest of this chapter easier to understand. A *set* is a collection of unique objects, called *elements*, or *members*. For example, the letters *A*, *B*, and *C* form a set of three letters. In mathematics (and in Python code syntax), sets are written inside curly braces, with the objects separated by commas: {*A*, *B*, *C*}.

Order doesn't matter for a set; the set {*A*, *B*, *C*} is the same set as {*C*, *B*, *A*}. Sets have distinct elements, meaning there are no duplicates: {*A*, *C*, *A*, *B*} has repeat *A*s and so is not a set.

A set is a *subset* of another set if it has only members of the other set. For example, {*A*, *C*} and {*B*, *C*} are both subsets of {*A*, *B*, *C*}, but {*A*, *C*, *D*} is not a subset of it. Conversely, {*A*, *B*, *C*} is a *superset* to {*A*, *C*} and also to {*B*, *C*} because it contains all their elements. The *empty set* { } is a set that contains no members at all. Empty sets are considered subsets of every possible set.

A subset can also include all the elements of the other set. For example, {*A*, *B*, *C*} is a subset of {*A*, *B*, *C*}. But a *proper subset*, or *strict subset*, is a subset that does not have all the set's elements. No set is a proper subset of itself: so {*A*, *B*, *C*} is a subset but not a proper subset of {*A*, *B*, *C*}. All other subsets are proper subsets. [Figure 6-1](#) shows a graphical representation of the set {*A*, *B*, *C*} and some of its subsets.